

KnockOnce

React Native Developer Intern - Take-Home Assignment

Remote | 3-Month Internship | Candidate evaluation draft

Objective	Assess React Native fundamentals, API integration, state management, mobile UX, debugging ability, and responsible AI-assisted development.
Expected effort	4-8 hours
Submission window	48 hours from receipt
Total marks	100

Problem Statement

Build a mobile application called **TaskFlow**. The app should demonstrate the ability to design a clean mobile interface, integrate with APIs, persist user session state, and structure code in a production-ready way.

Core Features

- Login with email and password validation.
- View tasks fetched from an API.
- Add a new task from the mobile UI.
- Mark tasks as completed or pending.
- Persist session using local storage.
- Include a basic profile or settings screen with logout.

Technical Requirements

Frontend	React Native; TypeScript preferred; React Navigation; Axios or Fetch; reusable components.
State Management	Context API, Redux Toolkit, or Zustand.
Backend / API	Use JSONPlaceholder or build a simple Node.js REST API with GET /tasks, POST /tasks, PATCH /tasks/:id.
Authentication	JWT simulation plus session persistence in AsyncStorage.
Code Quality	Loading states, empty states, error handling, folder structure, type safety, and meaningful commits.

Mandatory Screens

- **Login Screen** - email, password, validation, basic sign-in action.
- **Task Dashboard** - task list, pull-to-refresh, search, and completed/pending filters.
- **Add Task Screen** - create task form with validation.
- **Profile Screen** - user details and logout action.

Bonus Features

Bonus	Marks
Firebase Authentication	+5
Push Notifications	+5
Dark Mode	+3
Offline Task Caching	+5
App Store Ready Build	+5
Unit Testing	+5

AI-Assisted Development Requirement

Provide an **AI_USAGE.md** file describing which tools were used (for example Cursor, GitHub Copilot, ChatGPT, Claude, or Windsurf), which prompts were used, and which parts of the work were AI-generated versus manually written. Transparency matters.

Evaluation Rubric

React Native Fundamentals	20
API Integration	15
UI/UX Quality	10
State Management	10
Authentication Flow	10
Code Structure	10
Error Handling	5
Git Usage	5
AI Workflow Documentation	5
Bonus Features	10
Total	100

Submission Requirements

- GitHub repository with source code and a clear README.
- AI_USAGE.md with full disclosure of AI-assisted work.
- Short demo video (3-5 minutes) showing the app flow.
- APK or installable build for Android.

Shortlisting Guide

Score	Decision
90-100	Immediate final round

80-89	Technical discussion
70-79	Hold list
Below 70	Reject

Recommended Folder Structure

Path	Purpose
app/	Screens, navigation, and layout
components/	Reusable UI components
services/	API clients and helpers
store/	State management
utils/	Validation and shared utilities
assets/	Icons and images
README.md	Project overview and run instructions
AI_USAGE.md	AI tool usage disclosure

Submission Checklist

- App runs successfully on Android emulator or device.
- Login flow works and session persists after restart.
- Tasks can be fetched, added, searched, and updated.
- Code is readable, organized, and minimally documented.
- Demo video clearly explains architecture and feature choices.

This assignment is designed to identify candidates who can build, debug, and ship mobile features quickly while keeping the codebase maintainable and transparent.